Mamunul Karim

LinkedIn ResearchGate
Email: joymamunulkarim@gmail.com Phone: 307 761 9785
Address: Phoenix, 85021, AZ, USA

ACADEMIC CREDENTIALS

PhD (student) in Civil Engineering
Arizona State University
Expected Graduation: May 2029
Tempe, AZ, USA.
Start Date: Fall'26

Master of Science in Civil Engineering
University of Wyoming
CGPA: 3.67/4.00
Laramie, WY, USA.
Start Date: Fall'24
Courses: Pavement Management System, Pavement Design, Pavement Materials, Traffic Safety, Survey Construction, Introduction to Artificial Intelligence, Data Wrangling, and Spatial Data Science.
Major: Construction Engineering and Management (CEM)

B.Sc. in Building Engineering and Construction Management (BECM)
Khulna University of Engineering & Technology (KUET)
Equivalent to a US Civil Engineering BS degree
CGPA: 3.37/4.00; Final year Average GPA- 3.74/4.00
Major: Construction Engineering and Management (CEM)
Khulna-9203, Bangladesh.
February 2017 – July 2022

RESEARCH INTEREST

- Construction Cost Estimation, Digitization, Facility Management
- Infrastructure Asset Management, Workforce Development, Emerging Sensing Technologies
- Construction Technology: Automation, GenAI, Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Decision-Making in Construction
- Application of Virtual Reality (VR), Augmented Reality (AR), Mixed Reality in Construction, Application of UAV in Construction, Human-AI Collaboration, Visualization Technologies
- Building Information Modeling (BIM), Digital Twins, Internet-of-Things (IoT)
- Smart, Sustainable and Resilient Built Infrastructure, Sustainable Design, Sustainability

RESEARCH EXPERIENCE

- **PhD Research Direction**

Construction Workers' Safety (Underground Pipeline Safety, Mental Health, Workforce Development and Heat Stress)

- **MS Thesis**

Title: A Machine Learning and Statistical Analysis Framework for Enhanced Engineers' Estimate Accuracy in Highway Infrastructure Projects.
Project ID: CTIPS-015. Project Partners: USDOT, Office of the Assistant Secretary for Research and Technology; Wyoming Department of Transportation (WYDOT)

Conducted research on engineers' cost estimate accuracy using historical bid data from four U.S. states (Wyoming, Colorado, Utah, and Montana), applying K-means clustering to group projects and bid items by deviation level. Then, identified cost-significant bid items through Pareto analysis to determine the key items contributing to significant cost overruns, thereby improving estimation accuracy and financial transparency in highway infrastructure planning. Using the dataset, currently developing an LLM-based chatbot interface to help engineers and stakeholders quickly detect high-risk bids and reduce potential cost overruns through intuitive text-based interactions.

- **Undergraduate Thesis** *2021–2022*

Title: Automated Construction Material-Containing Vehicle Tracking and Inventory Management Using a Deep Learning Method.

This research presents an Automated Construction Material-Containing Vehicle Tracking and Inventory Management System utilizing YOLOv5-based Deep Learning. The system achieves 83% accuracy in vehicle detection, maintains real-time material updates, and calculates material quantities with 88% accuracy. These results demonstrate significant advancements in construction site logistics, promising enhanced efficiency and cost control. The findings underscore the system's potential to streamline operations and mitigate inventory discrepancies, offering valuable insights for future developments in automated construction management.

PUBLICATIONS

- **Karim, M.**, Ahmed, E., & Anzum, A.

"A comprehensive review and paradigm framework on Human – AI collaboration in construction management,"
International Conference on Industrialized Construction (IConIC 2026).

- **Karim, M.,** Abdelaty, A., & Yamany, M. S.

"Frequency-Aware Multi-Tier Machine Learning Framework for Item-Level Highway Construction Cost Estimation: An Improvement on Uniform Models"
International Journal of Construction Management

- Datta, S.D., Sahriea, F., Ashraf, J., **Karim, M.**, Shourav, M., Alam, M.T., Abdullah, M.S., Fuad, M., Mona, M.Z.H.

"Experimental and Machine Learning Analysis of Self-Compacting Concrete Made With Metalized Waste Plastic Fiber and Clay Brick Powder"
Journal of Engineering, 2026, 2822510, 22 pages, 2026. https://doi.org/10.1155/je/2822510

- **Karim, M.**, Mamun, M. A. A., Ira, R. T., & Mahamud, A., Islam, S. A., & Hossain, M. S.

"Literature Review on the Evolution of Analysis and Design of Pre-stressed Box Girder Bridge,"
International Journal on Science and Technology (IJSAT), Vol. 16, Issue 3.

- Khondoker, M. T. H., Datta, S.D., **Karim, M.,** & Joy, M. A. R.

"Forecasting Energy and Decarbonization: A Comprehensive BIM Simulation and Sensor Explainable AI Framework with Interactive Analytics"
Submitted to the Journal of Energy & Buildings. (Under review)

- Mahamud, A., Mamun, M. A., & **Karim, M.**

"A Simplified Stress-Crack Width Model for Strain Hardening Cementitious Material (SHCM),"
Journal of Construction and Building Materials Engineering (MAT Journals), Vol. 11, Issue 1 (Jan–Apr 2025), pp 47–58; DOI: https://doi.org/10.46610/JOCBME.2025.v011i01.005

- Mahamud, A., & Mamun, M. A., **Karim, M.**
“Effectiveness of Shear Walls According to the Position in a Structural Systems,” *Journal of Fareast International University,* Vol. 7, Issue 1, 2024, pp. 21–26.

GRANT & FELLOWSHIP

- **UTC Research Grant** *2024–2026*
Center for Transformative Infrastructure Preservation and Sustainability (CTIPS),
University of Wyoming.

Awarded a competitive research grant for the project ″A Machine Learning and Statistical Analysis Framework for Enhanced Engineer's Estimate Accuracy in Highway Infrastructure Projects″. The project addresses the challenge of significant deviations up to 25% between engineers’ cost estimates and awarded bids observed in Wyoming Department of Transportation (WYDOT) projects. It integrates quantitative analysis of historical bid tabulation data, survey-based evaluation of estimation practices, and advanced data-driven methods, including statistical modeling and machine learning, to improve cost estimation accuracy. Outcomes aim to reduce deviations in construction cost estimates between the engineer and the awarded bidder, enhance budget allocation efficiency, improve project performance, and promote transparency in transportation project planning and execution.

- **University Grants Commission (UGC) Research Grant** *2021-2022*
Khulna University of Engineering & Technology (KUET), Bangladesh.

Received competitive national-level funding as a Co-Investigator for a faculty-led project titled “Development of an Energy-Efficient HVAC System Using Earth-to-Air and Water Heat Exchanger.” Contributed to the experimental design, data acquisition, and performance evaluation of the HVAC system under Bangladesh’s climatic conditions. The project involved analyzing energy efficiency, optimizing system configuration, and assessing the integration of renewable energy principles to improve indoor thermal comfort and reduce operational costs in tropical environments.

STANDARDIZED TEST SCORE

- Graduate Record Examination (GRE) *March 2023*

Total Score	Verbal Reasoning	Quantitative Reasoning	Analytical Writing
304	143	161	3.0
Percentile	17th	65th	15th

SKILLS

- **Hard Skill**

Engineering & Design Software
AASHTOWare Pavement ME Design, AutoCAD, AutoCAD Civil 3D, Autodesk Navisworks Manage, Autodesk Design Review, Autodesk Revit, Autodesk Robot Structural Analysis, ETABS, SketchUp, Lumion, Microsoft Project, Primavera P6, Bluebeam Revu, Adobe Illustrator.

Programming & Statistical Data Analysis
Python, C++, R, IBM SPSS (Statistical Product and Service Solutions), Regression Models (Linear, Logistic, Quantile), Machine Learning and Deep Learning: Decision Trees, Random Forests, K-Means

Clustering, Gradient Boosting, XGBoost, Support Vector Machines (SVM), Convolutional Neural Networks (CNN), TensorFlow, and Keras, and Large Language Models (LLMs).

Geospatial & Remote Sensing
AutoCAD Map 3D, ArcGIS, ArcGIS Pro, QGIS, ESA SNAP, Metashape, CloudCompare, RealityCapture.

Office Productivity
Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Microsoft Publisher, Microsoft Project.

Others
Energy Simulation Tools: TRNSYS, EnergyPlus; Game Engine & Simulation: Unity Game Engine

- **Soft Skill**

Conflict Sensitivity, Leadership, Cross-cultural Communication, Collaboration and Teamwork, Problem-solving, Critical Thinking, and Decision Making.

ACADEMIC EXPERIENCE

- Graduate Research Associate *Fall 2026 – Present*

School of Sustainable Engineering and the Built Environment
Supervisor: Dr. Siyuan Song

- Conducted literature reviews on construction workers, mental health, underground pipeline safety, and workforce development.
- Proposed survey research and developed a survey questionnaire to inquire about the present situation of construction workers' mental health.
- Analyzed data on underground pipeline construction safety from 2000-2026 and investigated trends in accidents that occurred throughout the years.
- Compared underground pipeline accidents for construction workers before and after COVID.
- Generated ideas on analyzing data from construction sites for the heat stress monitoring study.

- Graduate Research Assistant (GRA) *Fall 2024 – Summer 2026*

Department of Civil and Architectural Engineering and Construction Management
University of Wyoming.
Supervisor: Dr. Ahmed Abdelaty
Project Title: A Machine Learning and Statistical Analysis Framework for Enhanced Engineer's Estimate Accuracy in Highway Infrastructure Projects, Phase I
Project ID: CTIPS-015
Project Partners: USDOT, Office of the Assistant Secretary for Research and Technology; Wyoming Department of Transportation

- Prepared and contributed to a grant proposal that was successfully funded, supporting research on Knowledge Management in Construction Projects.
- Led a data-driven investigation into cost estimation accuracy for more than 1500 of the latest highway infrastructure projects.

- Acquired and processed extensive historical bid tabulation datasets from Wyoming, Colorado, Montana, and Utah (2017–2024) by coordinating with the state DOT research offices.
- Computed average percentage deviations of 21.41%, 19.10%, 48.20%, and 24.00% for WYDOT, MDT, CDOT, and UDOT, respectively.
- Categorized projects into 4 different types (bridge, pavement, chip seal, miscellaneous) and classified them into high, moderate, and slight deviation clusters using K-means clustering.
- Conducted statistical analyses (Modified Z score, ANOVA test)to identify estimation patterns and correlations, and performed Pareto analysis to determine the costliest and most significant deviated bid items.
- Integrated clustering outputs with project-level deviation metrics to reveal systemic under- and over-estimation trends.
- Designed and prepared survey instruments targeting engineers and project managers to assess organizational estimating practices.
- Supported literature review efforts to frame the study within national and state-level contexts.
- Developed high-quality visualizations to communicate research findings.
- Coordinated with project partners to ensure research alignment and goal achievement.

o Research Assistant (RA) *July 2022- February 2023*

Department of Building Engineering and Construction Management (BECM),
Khulna University of Engineering & Technology (KUET), Bangladesh
Title: Current Scenario of Understanding and Adaptation of Sustainable Building Practices in Bangladesh
Supervisor: Md. Jewel Rana, Assistant Professor, BECM

- Conducted an extensive study on the current state of sustainable building practices in Bangladesh's construction industry, focusing on barriers and stakeholder perceptions.
- Performed a comprehensive literature review covering the evolution of sustainable development, global and national policies, and sustainable construction principles.
- Designed and distributed online and offline survey instruments targeting engineers, contractors, architects, owners, CEOs, and academicians.
- Collected and analyzed over 100 stakeholder responses using quantitative statistical methods, including Reliability Index (RII), Cronbach's Alpha, and Kruskal-Wallis tests.
- Collaborated with faculty and peers to validate survey instruments and ensure methodological soundness for academic and field-level relevance.
- Identified key challenges such as lack of awareness, high costs, inadequate training, and absence of regulatory guidelines affecting sustainable construction adoption.
- Proposed a sustainable project management framework based on stakeholder feedback and international best practices.

o Teaching Assistant (TA) *March 2023- August 2023*

Trained by: International Teaching Assistantship (ITA) Workshop – University of Wyoming

Department of BECM, KUET, Bangladesh.
Courses Assisted:
Sessional on Building & Construction Materials, Construction Estimating, Construction Techniques and Equipment, Sessional on Building Information Modeling (BIM)

- Conducted lab demonstrations and supported students during material testing experiments using standard equipment and procedures.
- Reviewed structural designs and assisted students in preparing comprehensive construction cost estimates.
- Assisted in creating detailed drawings for high-rise buildings, including floor plans, parking layouts, and wall sections.
- Led site visits and field-based instruction to introduce students to practical construction techniques used across Bangladesh.
- Provided hands-on support with BIM software, including Autodesk AutoCAD, Autodesk Revit, Autodesk Navisworks Manage, and Primavera P6.
- Facilitated integration of multiple software platforms to demonstrate collaborative modeling and project scheduling workflows.
- Assisted faculty in evaluating student performance by grading assignments, providing constructive feedback, and maintaining academic records to support continuous improvement.

PROFESSIONAL EXPERIENCE

- **Project Engineer** *February 2024- July 2024*
  Bang Jin Group

  - Provided technical oversight on materials and geotechnical testing projects related to soil compaction and settlement.
  - Successfully introduced and initiated the manufacturing of Prefabricated Vertical Drains (PVD) for the first time in Bangladesh.
  - Offered expert consultancy on selecting optimal geotechnical and construction materials.
  - Conducted site visits and liaised with BWDB (Bangladesh Water Development Board), RHD (Roads and Highways Department), LGED (Local Government Engineering Department), and PWD (Public Works Department) officials.

- **Assistant Project Engineer** *September 2023- January 2024*
  Aysha Construction and Suppliers

  - Performed structural design calculations, detailing, and review of building elements.
  - Interpreted and validated technical documents and engineering drawings.

- **Intern** *January 2022- June 2022*
  United Purpose (BIM Modeler) & Amigos Construction

  - Worked under the supervision of Assistant Project Engineer Maria Binte Mannan.
  - Created a 3D hilly area landscape model and a rural power plant model using Revit and Lumion.
  - Supported planning, visualization, and documentation for development projects.
  - Supervised pile casting activities and performed quality control of materials.
  - Maintained safety protocols and worked on 2D/3D drawings and BIM modeling documentation.

- **Science Tutor** *January 2018- December 2021*
  Caretutors
  - Tutored high school students in mathematics, physics, and chemistry.
  - Consistently maintained a 5.0 out of 5.0 tutor rating based on student and parent feedback.

ACADEMIC PROJECTS EXPERIENCE

- **AASHTOWare Pavement ME Design (Software): A Case Study of Champaign, IL.**

Pavement Design *Spring 2025*

  - The MEPDG framework was applied in this software to design flexible and rigid pavements for interstate, state, and local roads in Champaign, Illinois.
  - This software was used to simulate performance under local traffic and climate conditions.
  - Optimal pavement configurations were identified based on distress thresholds, cost-effectiveness, and structural durability.
  - Conducted sensitivity analysis to evaluate the impact of varying input parameters on pavement performance predictions, ensuring robust and reliable design outcomes.

- **Utilization of Bottom Ash in Asphalt Mixes**

Pavement Materials *Fall 2024*

  - Bottom ash from three Wyoming power plants was incorporated into asphalt mixes using the Marshall and SuperPave methods.
  - Laboratory tests, including TSRST and GLWT, were conducted to evaluate the resistance to low-temperature cracking and rutting.
  - Bottom ash mixes were found to be feasible with proper gradation and asphalt content adjustments.

- **AI-Based Smart Security System for Edge Devices**

Introduction to Artificial Intelligence *Fall 2024*

  - MobileFaceNet and Dlib models were fine-tuned and evaluated using the CASIA-WebFace dataset.
  - An MDP-based decision-making system was integrated for adaptive, real-time responses on edge devices.
  - System performance was assessed in terms of accuracy, latency, and resource efficiency.

- **Geospatial and Statistical Analysis of Construction Bid Deviations in Rocky Mountain Regions.**

Basics of Spatial Data Science *Spring 2025*

  - Public bid data from Rocky Mountain cities was analysed using R to examine construction bid deviations.
  - Spatial and multivariate statistical analyses were conducted to assess the influence of geographic, demographic, and project-related variables.

- **Statewide High School Performance Analysis in R: A Comprehensive Data Integration and Statistical Profiling of Standardized Test Scores by County.**

Data Wrangling *Fall 2024*

- All county-level datasets were merged, cleaned, and saved as a unified RDS file using a reproducible R script.
- Summary statistics and categorical counts were generated to validate and explore the cleaned dataset.
- Multiple analytical tables were created and exported to CSVs, using R for statistical summarization and data reshaping.

PROFESSIONAL PROJECTS EXPERIENCE

- **Implementation of PVD for Soil Compaction (Bang Jin Group)**

Provided technical direction for geotechnical testing and construction projects focused on soil compaction and settlement. Spearheaded the introduction of Prefabricated Vertical Drains (PVD) in Bangladesh for the first time, facilitating more effective ground improvement. Collaborated with local government agencies, including BWDB, LGED, and PWD, for material approval and project rollout.

- **Structural Design of Residential and Commercial Elements (Aysha Construction and Suppliers)**

Performed load-bearing calculations, member sizing, and structural detailing for mid-rise residential structures. Reviewed structural plans and technical documents to ensure accuracy and regulatory compliance. Supported cross-functional collaboration with architects for construction documentation.

- **3D BIM Modeling for Sustainable Development Projects (United Purpose)**

Created detailed 3D BIM models of rural infrastructure using Autodesk Revit and Lumion under the supervision of the Assistant Project Engineer. Developed visualizations for a hilly landscape area and a solar-powered rural facility, aligning with sustainable development goals. Delivered complete design packages, including renders and planning layouts, for presentation to stakeholders.

ORGANIZATIONAL AND MANAGERIAL EXPERIENCE

- President *October 2025- Present*

Uwyo ITE Student Chapter, University of Wyoming

- Communication Secretary *November 2024- Present*

Bangladesh Students' Association (BSA), University of Wyoming

- Vice President *January 2018- June 2018*

BECM Association, KUET

- General Secretary *July 2018- December 2018*

KUET Math Club

ACCOLADES

- Awarded Fulton Fellowship by Arizona State University for academic research excellence.
- Elected as President of the ITE - University of Wyoming Student Chapter.
- Awarded the 2025 ITE (Institute of Transportation Engineers) CO/WY Section Graduate Level Scholarship by the ITE Mountain District.
- Awarded an International Scholarship by the University of Wyoming for excellence and leadership.
- Received a 4-Year Bangladesh Technical Scholarship for consistent academic performance during

the B.Sc. program.

- Awarded 2-Year Bangladesh Government Merit Scholarship for securing the highest academic grade
- Champion in Poster Presentation on *"Green Building Adaptation in the Context of Bangladesh"*, Builtech Fest, KUET.
- Runner-up in Case Study Competition on *"Construction Material Detection at Under-Construction Sites Using YOLOv5"*, Builtech Fest, KUET.

CERTIFICATES

Workshop on Construction Management, ITA Training Workshops, Responsible Conduct of Research for Engineers, and IRB Member (Human Subjects Research).